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Accurate long-term prognostics of proton exchange membrane fuel cells using recurrent and convolutional neural networks

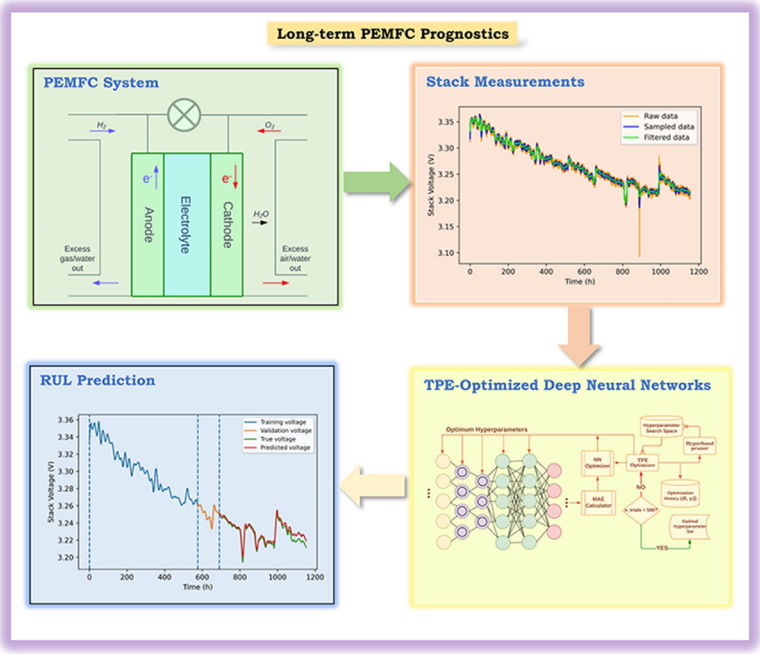
Kartik Sahajpal[*], *K.P.S. Rana*, *Vineet Kumar*

Instrumentation and Control Engineering Department, Netaji Subhas University of Technology, Sector-3, Dwarka, New Delhi, 110078, India

HIGHLIGHTS

- Six deep learning models are developed for PEMFC RUL and degradation prediction.
- Hyperparameters of each model are optimized using TPE for static and dynamic loading.
- Static loading models exhibited excellent transfer learning performance.
- All tuned models performed superior to recent reported approaches in RMSE.
- LSTM and 1D–CNN–Bidirectional-GRU demonstrated best results trained on 50% data.

GRAPHICAL ABSTRACT



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ABSTRACT

Estimating the remaining useful life (RUL) of proton exchange membrane fuel cells (PEMFCs) is key to predicting and increasing their durability. This paper investigates six deep learning techniques – long short-term memory (LSTM) networks, gated recurrent

* *Corresponding author*. Instrumentation and Control Engineering Department, Netaji Subhas University of Technology, Sector-3, Dwarka, New Delhi, 110078, India.
E-mail addresses: kartik.sahajpal.ug20@nsut.ac.in (K. Sahajpal), kpsrana@nsut.ac.in (K.P.S. Rana), vineet.kumar@nsut.ac.in (V. Kumar).
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units (GRU), 1-D convolutional neural network (CNN)-LSTMs, 1D–CNN–GRUs, 1D–CNN–Bidirectional-LSTMs, and 1D–CNN–Bidirectional-GRUs – for RUL and long-term degradation trend prediction for a five-cell PEMFC stack operated with and without current ripples. The hyperparameters of all models are optimized for minimum root-mean-square error (RMSE) of the stack voltage using tree-structured Parzen estimators (TPE) with hyperband pruning. The obtained prediction results are benchmarked using the coefficient of determination (R^2), RMSE, and mean absolute percentage error (MAPE) in stack voltage predictions and relative error (RE) in RUL estimates. The comparative analysis for models trained on 50% datasets reveals that the best R^2, RMSE, MAPE, and RE of 98.87E-2, 1.258E-3, 0.285E-1, and 0.0425E-1 are offered by the LSTM model for the stack without current ripples. Further, for the stack with current ripples, the best R^2, RMSE, MAPE, and RE of 99.482E-2, 1.77E-3, 0.444E-1, and 0.089E-1 are exhibited by the 1D–CNN–Bidirectional-GRU model. The LSTM model also reveals excellent transfer learning performance on the stack with current ripples with the best R^2, RMSE, MAPE, and RE of 98.91E-2, 2.455E-3, 0.362E-1, and 0.044E-1, respectively.

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Introduction

Proton exchange membrane fuel cells (PEMFCs) are potential zero-emission commercial renewable energy sources for electric vehicles [1] and power generation plants [2]. PEMFCs are free from moving parts and hence show higher energy conversion efficiency, quicker startup, and lower acoustic noise than their fossil-fuel counterparts. PEMFCs generate power through the electrochemical reaction of hydrogen with oxygen over a platinum catalyst, as shown in Fig. 1. However, their applications are currently small-scale and limited to the military and automotive industries due to their short useful life (~1500–2000 h) [3] and high cost [4]. In these applications, intelligent and adaptive control schemes need to be designed to adjust input variables such as gas flow rates to maximize the efficient and stable operation of PEMFC systems [5–7]. For large-scale industrial applications such as electric vehicles, a useful life of at least 6000 h is required [8]. Recent research reveals that estimating the mean time to failure or the remaining useful life (RUL) can help select maintenance actions and optimal operating conditions beforehand to increase the durability of PEMFCs [8,9]. This helps further minimize unpredictable failures and unscheduled maintenance, which decreases resource consumption, environmental waste, and maintenance costs. RUL estimation or prognostics also allows for predicting the future state of health of the PEMFC without physically operating it for a long time, further saving time and costs. Therefore, combined with post-prognostics decision-making, RUL estimation is a powerful technique to help maximize the service life of PEMFC-powered systems.

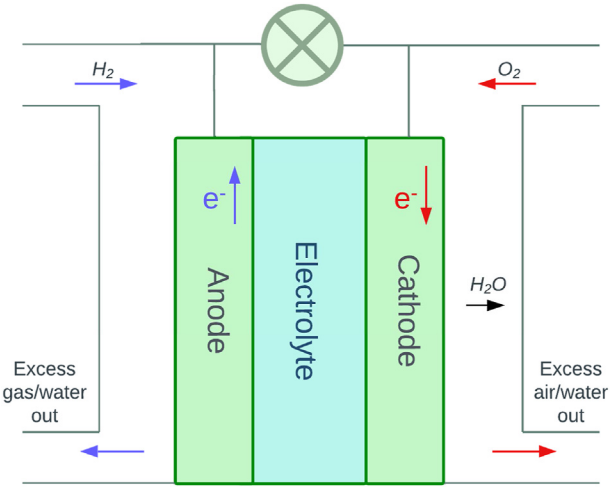


Fig. 1 – A simplified operational diagram of a PEMFC.

The RUL of PEMFCs depends on the output voltage and power degradation dynamics, which further depend on various operational factors, such as loading conditions and usage application. According to Pei et al. [10], the significant operational conditions of a PEMFC are start-stop cycles, high-power loading, load-changing cycles, and idle conditions. Their analysis shows that load-changing (or dynamic loading) and start-stop cycling contribute to 56.5% and 33% of the PEMFC voltage degradation, respectively. Further, the effects of dynamic loading and start-stop cycling on PEMFC voltage degradation have also been analyzed in Refs. [11,12]. Studies show that catalyst degradation, thermal runaway due to poor thermal management, membrane thinning and decomposition, carbon corrosion, catalyst poisoning, and diffusion stress significantly contribute to the degradation of PEMFCs under start-stop and dynamic loading [13]. Several approaches have been reviewed to model various degradation mechanisms in PEMFC-powered systems [14]. But regardless of the modeling approach, a degradation indicator that correlates well with the extent of degradation of the PEMFC is necessary to estimate its RUL quantitatively [13,14]. The most used indicators to quantify the extent of PEMFC degradation are its output voltage and power. Still, some studies have also used electrochemically active surface area [15] or a combination of degradation indicators [16] to estimate the RUL of PEMFCs.

Estimating the RUL of a PEMFC-powered system requires extensive temporal measurements of several parameters of the PEMFC to design, validate, and test the predictive performance of RUL estimators. Recent PEMFC prognostics literature

Table 1 – Measurements in the IEEE PHM 2014 data challenge.

Model inputs	Physical significance
U1 to U5	Single-cell voltage (V)
Utot	Stack voltage (V)
I	Current (A)
J	Current density (A/cm^2)
TinH2	Inlet temperature of H_2 (°C)
ToutH2	Outlet temperature of H_2 (°C)
TinAIR	Inlet temperature of Air (°C)
ToutAIR	Outlet temperature of Air (°C)
TinWAT	Inlet temperature of cooling Water (°C)
ToutWAT	Outlet temperature of cooling Water (°C)
PinH2	Inlet Pressure of H_2 (millibar absolute, or mbara)
PoutH2	Outlet Pressure of H_2 (mbara)
PinAIR	Inlet Pressure of Air (mbara)
PoutAIR	Outlet Pressure of Air (mbara)
DinH2	Inlet flow rate of H_2 (L/min)
DoutH2	Outlet flow rate of H_2 (L/min)
DinAIR	Inlet flow rate of Air (L/min)
DoutAIR	Outlet flow rate of Air (L/min)
DWAT	Flow rate of cooling water (L/min)
HrAIRFC	Inlet Hygrometry (Air) - estimated (%)

is primarily based on the publicly available IEEE 2014 Prognostics and Health Management (PHM) Data Challenge Dataset provided by FCLAB Research Federation, France [17]. This dataset was derived from around 7-week-long experiments with two 5-cell PEMFC stacks on the FR CNRS 3539 testbench at FCLAB Research Federation, France. The various PEMFC parameters measured, including the cell and stack voltages and currents and the gas flow rate, temperature, and humidity, are summarized in Table 1. The first stack, designated as FC1, was operated for 1155 h at 70 A load current. The second stack, designated as FC2, was operated for 1021 h at 70 A load current with 7 A, 5 kHz triangular current ripples. The current ripples in FC2 emulate a power converter connected to the stack output and represent the dynamic or quasi-dynamic operation of the PEMFC. Each cell in both stacks had an active area of 100 cm^2, a nominal current density of 0.70 A/cm^2, and a maximal current density of 1 A/cm^2.

Only a few long-term measurement experiments other than that of the IEEE PHM 2014 Data Challenge Datasets have been reported, and not all are made publicly available. For instance, Zuo et al. [18] conducted a 1008-h dynamic durability test on a PEMFC cell. They built deep learning prognostic models with gated recurrent units (GRUs) and long short-term memory (LSTM) networks and evaluated the predictive performance of the recurrent neural networks (RNNs) with and without attention mechanisms. Another study conducted a 5000-h durability test under variable ambient temperature on a twenty-eight-cell open-cathode PEMFC system used in an electric vehicle [19]. An ESN-based degradation prediction model was validated on the resulting dataset and exhibited a normalized RMSE of 0.098 over a prediction timeframe of over 2000 h.

From the literature, three approaches to PEMFC prognostics can be identified – model-based, data-driven, and hybrid methods [8,9]. Model-based approaches use physical or empirical models that describe the PEMFC dynamics and degradation via a set of algebraic and/or integrodifferential equations. A significant advantage of model-based approaches is that they do not require extensive experimental measurements or empirical data. For instance, Zhang et al. [20] proposed an ensemble of two physics-based degradation models based on the stack voltage and state of health as degradation indicators for FC1. The degradation of the PEMFC's internal resistance and open-circuit voltage estimated the state of health, while the RUL was predicted by joint state-parameter estimation using a particle filter. Chen et al. [21] proposed three voltage degradation models and an unscented Kalman filter to predict the degradation of field-operated PEMFCs powering postal electric vehicles. In another approach, a hybrid indicator, consisting of the stack voltage, power and internal resistance of the PEMFC, was used to predict the degradation trend of FC1 using a second-order Gaussian degradation model and an unscented particle filter [22]. However, accurate degradation models are challenging to develop as the exact degradation mechanisms of PEMFCs are not entirely understood and the degradation occurs in multiple temporal and dimensional scales. Notably, model-based prognostics approaches often make assumptions about the underlying physics of the system and the nature of the degradation process. If these assumptions are incorrect or incomplete, the model may not accurately predict the RUL and degradation trend of the system. Moreover, model-based approaches are usually computationally complex and unsuitable for online RUL prediction [9].

Data-driven approaches use measurements on the PEMFC to build black box models for the degradation dynamics of PEMFCs. Data-driven approaches such as machine learning and deep learning do not require a physical or empirical model of PEMFC systems. Hence, they can easily model non-linear characteristics and accurately predict degradation trends from input-output data without making assumptions about the degradation dynamics of the PEMFC system. For instance, LSTM networks have shown excellent prediction performance on multivariate time series datasets, including FC1 and FC2 [23–26]. Zuo et al. [23] proposed an LSTM network degradation model to predict the degradation trend of FC1 after Savitzky-Golay smoothing and validated the model's performance on FC2. In another study, LSTM and bidirectional LSTM models with attention mechanisms [24] were trained on FC1 and FC2 after random forest feature extraction to achieve a relative prediction error of around 0.09%–0.29%. Bidirectional LSTMs have also been combined with GRUs with an echo state network (ESN) for short-term degradation prediction using only one-fifth of the FC1 and FC2 datasets for model training [25]. Wang et al. [26] proposed an ensemble of stacked LSTM with attention, support vector regression and random forest – AT-LSTM-SVR-RF – optimized using differential evolution for RUL and long-term degradation prediction of FC1 and FC2.

In addition to RNNs, convolutional neural networks (CNNs) have also gained increasing attention in the prognostics literature due to their ability to capture complex degradation patterns in multivariate time series. For instance, a CNN prognostic model trained on features selected by a random forest algorithm has been shown by Huo et al. [27] to predict the

I–V polarization curves of PEMFCs based on a dataset of 140 I–V polarization curves. Benaggoune et al. [28] reported high long-term degradation trend prediction accuracy on FC1 and FC2 with an attention-based dilated CNN for multi-step-ahead prediction and a conditional CNN for long-term prognostics. More recently, RNNs and CNNs have been combined to achieve better accuracy in long-term degradation trend prediction than of RNNs and CNNs alone. For instance Ref. [29], demonstrated a CNN-Bidirectional-LSTM network architecture for long-term degradation trend prediction and RUL estimation of FC1 and FC2. Another deep learning architecture with a two-dimensional CNN and a GRU net optimized by a manta ray foraging optimizer was reported by Tao et al. [30] for performance degradation prediction of a PEMFC system.

Hybrid approaches fuse model-based with data-driven methods to achieve better prediction accuracy and have only recently gained research interest in RUL estimation. Cheng et al. [31] proposed a hybrid prognostic methodology with a data-driven least square support vector machine model and a model-based regularized particle filter approach to estimate the RUL of FC1 and its probability distribution. A hybrid prognostic model [32] with an adaptive unscented Kalman filter for aging state estimation and a particle swarm optimization (PSO)-adaptive neuro-fuzzy inference system framework has also been explored for long-term degradation prediction of FC1 and FC2. Adaptive Kalman filters have also been combined with an empirical aging model hybridized with a non-linear autoregressive exogenous model to predict the reversible and irreversible degradation trends for FC1 and FC2 [33]. Both [32,33] investigate the effect of varying training set sizes on the prediction performance of the hybrid models. But hybrid approaches require experimental data and knowledge of the physical degradation laws. Also, hybrid models often cumulate the disadvantages and advantages of data-driven and model-based approaches [9]. Detailed reviews of various approaches to PEMFC RUL estimation have been reported in Refs. [34,35].

Deep learning is a data-driven prediction technique that has gained increasing attention for reliability analysis, prognostics, and health management of systems prone to operational failure [36,37]. Neural networks with at least two layers, or deep neural networks, mathematically model human neurons and emulate the human learning process in two phases. In the context of PEMFC RUL prognostics, the first phase is the learning phase, where the features (here, measurements) recorded before the prediction interval are used to train the neural network so that the network updates its parameters to best-fit the degradation behavior of PEMFCs. In the prediction phase, the neural network predicts the future degradation trend of PEMFCs using the learned weights at times greater than the prediction time. Wang et al. [38] proposed an LSTM model driven by a navigation sequence generated by an autoregressive integrated moving average model with exogenous variables for a 96-cell PEMFC stack operated under static loading and a 15-cell PEMFC stack used under dynamic loading. Meraghni et al. [39] proposed a digital twin stacked autoencoder for RUL and degradation trend prediction of a five-cell PEMFC stack with voltage measurements over 1750 h. Using the stack voltage, current, temperature and reactant pressure measurements, a multi-input ESN has also been reported [40] to predict the RUL of FC1 and FC2.

The predictive performance of deep learning models depends on several hyperparameters that determine how their predictions generalize to new, unseen data. The hyperparameters generally include the neural network activation function and the number of neurons and hidden layers. It may be noted that manually tuning these for optimal prediction results is not practically feasible, as the exact relationships between the values of a typical hyperparameter and a model's prediction accuracy are poorly understood. To achieve optimum predictive performance, a set of hyperparameters that minimizes the desired cost function needs to be searched. This forms a multivariable optimization problem addressed in the PEMFC prognostics literature using conventional grid search techniques, meta-heuristic approaches such as evolutionary algorithms and swarm intelligence techniques, or sequential design approaches. Grid search [18] evaluates each possible combination of hyperparameters in the search space to find the best model. Therefore, it is computationally inefficient, especially when a neural network takes a long time to train. Additionally, it does not propagate information between evaluation trials to eliminate areas of the hyperparameter space that show poor predictive performance. On the other hand, meta-heuristic algorithms, such as genetic algorithms, PSO, and gray-wolf optimizer (GWO), direct the search process to generate, tune, or find heuristics that minimize the desired cost function [26,30,31,37]. Chen et al. [41] reported a backpropagation neural network with hyperparameters optimized by the mind evolutionary algorithm, PSO, and a genetic algorithm for the aging prognosis of FC1. Li et al. employed an improved GWO algorithm to optimize the hyperparameters of a kernel extreme learning machine predictor during the validation phase [42]. However, if misguided, population-search algorithms may grow exponentially in computational costs as the number of objectives and the dimensions of the search space increase [43]. An intelligent hyperparameter selector, such as one based on Bayesian optimization [44], would focus on only those regions that optimize an objective function, such as those that minimize the RMSE or MAPE, with reduced computational effort. Bayesian optimizers have demonstrated excellent degradation prediction performance on Li-Ion batteries [45,46] and bearings used in rotating machinery [47].

Based on the above literature review, it can be inferred that deep learning techniques are promising candidates for RUL and long-term degradation prediction of PEMFCs. Hence, motivated by the above observations, a deep learning-based approach was explored with tree-structured Parzen estimators (TPE) – a modified Bayesian optimizer – for hyperparameter tuning to develop a degradation prediction and RUL estimation model for PEMFCs under static and dynamic loading conditions. The presence and absence of current ripples emulated the static and dynamic loading conditions. The proposed PEMFC degradation models are developed and validated on the two IEEE PHM 2014 Data Challenge Datasets – FC1 and FC2. The efficiency of six deep neural network models with recurrent and convolutional layers is explored for estimating the remaining useful life of a five-cell PEMFC stack. The PEMFC

stack voltage is selected as the degradation indicator. The main contributions of this paper can be summarized as follows.

- Six deep learning models for predicting the RUL of PEMFCs under static and dynamic loading conditions – LSTM, 1D–CNN–LSTM, 1D–CNN–Bidirectional-LSTM, GRU, 1D–CNN–GRU, and 1D–CNN–Bidirectional-GRU – are investigated.
- Hyperparameter optimization is carried out using tree-structured Parzen estimators (TPE) and hyperband pruning to get the best performance of each model under the two loading conditions.
- Performance analysis of the models against various hybrid and purely recurrent and convolutional neural networks for RUL estimation is presented.
- Prediction results show that all deep learning models proposed in this work show superior degradation trend and RUL prediction performance to deep learning models reported in the existing literature.
- Each deep learning model trained on the FC1 dataset could accurately predict the degradation trend and RUL of FC2.
- The proposed prognostic approach is suitable for online RUL estimation and long-term degradation trend prediction of PEMFC-powered devices.

This paper is organized as follows. Section 2 presents the model architectures and the loss functions used to benchmark the performance of the proposed six prognostic models. Section 3 briefs the implementation of the proposed prognostics strategy. Section 4 analyzes the long-term degradation trend and RUL prediction results of the six models trained on the FC1 and FC2 datasets and the transfer learning performance of the six models trained on the FC1 dataset. Section 5 summarizes the findings and provides future directions for research.

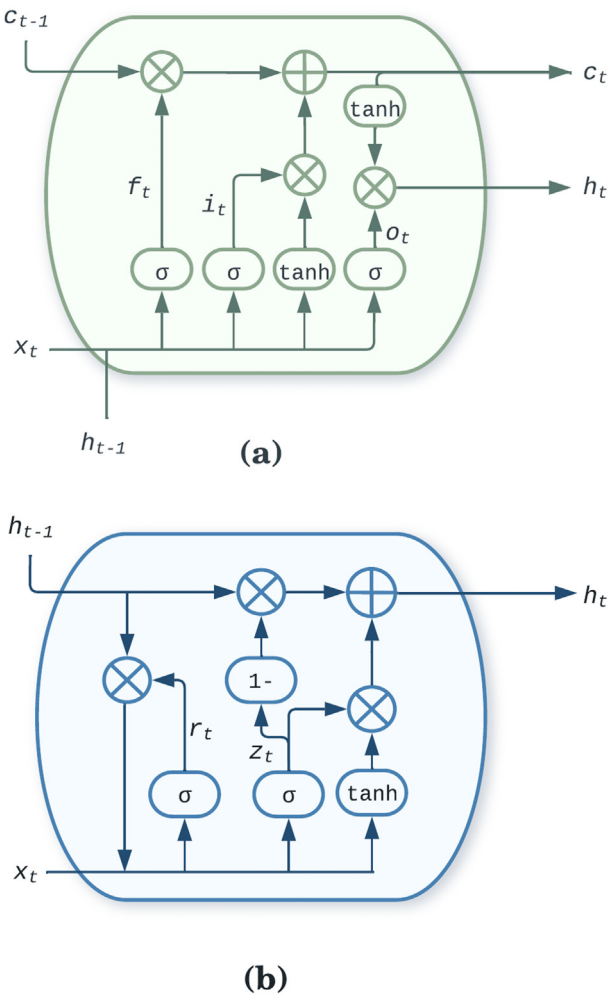


Fig. 2 – (a) Long short-term memory cell; (b) Gated recurrent unit.

Deep learning-based degradation prediction models

This section presents the architectures of the developed deep learning models and briefly describes the TPE optimizer.

Deep learning architectures

This section introduces the two investigated neural network architectures – RNNs and CNNs.

Recurrent neural networks

RNNs can model fixed and variable-length sequences such as speech and time series using internal state variables as memory. The LSTM network is an improved RNN architecture that solves the vanishing gradient and gradient explosion problems in conventional RNNs. LSTMs have found wide applications in speech, text, natural language processing, and time series forecasting [48]. LSTMs consist of memory blocks rather than neurons as units, as shown in Fig. 2(a). Each memory block has one or more cells that contain an input gate i_t, a forget gate f_t, and an output gate o_t. The internal state of an LSTM cell at time t is c_t. The forget gate forgets irrelevant training data from previous timesteps.

The governing equations of the LSTM are given by:

$$i_t = \sigma(W_{xi}\, x_t + W_{hi}\, h_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_{xf}\, x_t + W_{hf}\, h_{t-1} + b_f) \quad (2)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_{xc}\, x_t + W_{hc}\, h_{t-1} + b_c) \quad (3)$$

$$o_t = \sigma(W_{xo}\, x_t + W_{ho}\, h_{t-1} + b_o) \quad (4)$$

$$h_t = o_t \odot \tanh(c_t) \quad (5)$$

Here, W_{xi}, W_{xf}, W_{xc}, and W_{xo} are the weight matrices connecting x_t to i_t, f_t, c_t, and o_t, respectively; W_{hi}, W_{hf}, W_{hc}, and W_{ho} are the weight matrices connecting h_{t-1} to i_t, f_t, c_t, and o_t, respectively; and b_i, b_f, b_c, and b_o are the respective bias weights. $\sigma(\cdot)$ represents the sigmoidal activation function and $\odot$ denotes the elementwise or Hadamard product operator.

The GRU [49] is an RNN slightly different from LSTM in that each cell in a memory block has an update gate, a reset gate,

and an output gate. The input gate and forget gate of an LSTM are combined in function into the update gate of the GRU, as shown in Fig. 2(b). The reset gate determines whether the past hidden state is to be ignored.

The equations governing the outputs of the update gate z_t and the reset gate r_t are given by Ref. [50]:

$$z_t = \sigma(W_{xz}\, x_t + W_{hz}\, h_{t-1} + b_z) \tag{6}$$

$$r_t = \sigma(W_{xr}\, x_t + W_{hr}\, h_{t-1} + b_r) \tag{7}$$

The candidate activation vector $\tilde{h}_t$ and the hidden layer output h_t are given by:

$$\tilde{h}_t = tanh(W_{xh} x_t + W_{hh}\, (r_t \odot h_{t-1}) + b_h) \tag{8}$$

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t \tag{9}$$

Here, W_{xz}, W_{xr}, and W_{xh} are the weight matrices connecting x_t to z_t, r_t, and $\tilde{h}_t$, respectively; W_{hz}, W_{hr}, and W_{hh} are the weight matrices connecting h_{t-1} to z_t, r_t, and $\tilde{h}_t$, respectively; b_z, b_r, and b_h are the respective bias weights.

The LSTM and GRU networks with the cells shown in Fig. 2 process the input sequence only in the forward direction. Partitioning the neurons in the hidden layer into two sets – one that processes the sequence in the forward direction and the other that processes the sequence in the backward direction – results in a bidirectional recurrent neural network. Connecting these two sets of forward and backward processing networks to the same output layer may improve prediction performance. The bidirectional model can leverage previous and past contexts to learn long-term dependencies and bidirectional trends. Hence, bidirectional LSTMs [51] and GRUs [52] have recently found increasing interest in prognostics and predictive maintenance applications.

Convolutional neural networks

CNNs process data with a grid-like structure, such as time series and images. A typical CNN consists of a convolutional layer, a pooling layer, and an output layer. The convolutional layers extract feature maps from the input data by convolving the input data, x, with learnable fixed-length kernels (or filters), k, that slide over the input data with a preset stride. These layers compute the dot product of a portion of the input data with an overlapped filter of the same size. The results of this convolution are then propagated through a non-linear activation function, f, to generate the output feature map for the kth layer h_k:

$$h_k = f(w_k * x + b_k) \tag{10}$$

Here, w_k and b_k are the convolutional layer's filter weights and biases, respectively.

The convolutional layer may be followed by a pooling layer, especially in computer vision applications, to downsample feature maps by filtering the maximum (or mean) values of each filter operation. Pooling may lower training time per epoch and help avoid overfitting by focusing on dominant features. However, it may show poor results on datasets with minute and irregular key features. Also, pooling makes the CNN learn only those features which are position- or translation-invariant and thus may discard important temporal information in the features [53]. In this work, CNNs lack pooling layers, and models with CNN and recurrent layers contain CNN layers connected to recurrent layers by feeding the output of the convolutional layer to the recurrent layer.

Hyperparameter optimization using tree-structured Parzen estimators and hyperband pruning

The TPE algorithm is a modified Bayesian optimizer recently reported for hyperparameter optimization of long-term prognostic models for steam generators in nuclear power plants [54]. Bayesian optimization uses a Gaussian process model of the objective function and is especially useful when the objective function f is computationally expensive to evaluate. Consider a history of hyperparameter sets (θ_i) and the objective function evaluated at each hyperparameter set (y_i) bundled as a vector $X = \{(\theta_o, y_o), \ldots, (\theta_{n-1}, y_{n-1})\}$. Bayesian optimization models f as $p(X)$ and selects the next hyperparameter set using an acquisition function that balances sampling from areas with high randomness (exploration) with that from areas with optimal value (exploitation) in the hyperparameter search space.

Bayesian optimization then iteratively samples the next θ that maximizes $a(\theta)$, i.e., $\theta_{i+1} = argmax_{\theta \in Z}\, a(\theta)$, evaluates $y_{i+1} = f(\theta_{i+1})$, and appends (θ_{i+1}, y_{i+1}) to X. The probabilistic model $p(f|X)$ is then updated after each sampling trial. The iteration continues until a stopping criterion, such as a fixed number of trials or a time limit, is met.

The TPE algorithm uses kernel averages centered on existing data points to estimate the probability density of the Gaussian process model of f, rather than evaluating $p(X)$ [55]. These kernel density estimators, associated with optimal and poor hyperparameter sets, respectively, are given by:

$$l(\theta) = p(y < y^* | \theta, X) \tag{11}$$

$$g(\theta) = p(y > y^* | \theta, X) \tag{12}$$

The pseudocode for the TPE algorithm with hyperband pruning is shown in Fig. 3. Here, M_i is a computationally cheaper proxy function for the objective function f fitted in the ith trial, N is the number of trials, and θ^* is the hyperparameter set that minimizes the acquisition function, which is the expected improvement (*EI*).

The *EI* is the expectation under M of f that $f(\theta)$ negatively exceeds a threshold y^*:

$$EI_{y^*}(\theta) = \int_{-\infty}^{y^*} (y - y^*) p(y|\theta) dy \tag{13}$$

$$EI_{y^*}(\theta) = \int_{-\infty}^{y^*} (y - y^*) \frac{p(\theta|y) p(y)}{p(\theta)} dy \tag{14}$$

TPE chooses $\gamma = p(y < y^*)$. Now,

$$p(\theta) = \int_R p(\theta|y) p(y) dy = \gamma\, l(\theta) + (1 - \gamma) g(\theta).$$

The right-hand side of the integral can be expressed as:

```
TPE_Hyperband(f, M, N)
    H ← ∅
    for i ← 1 to N
        θ* ← argmin_θ (EI(θ, M_{i−1}))
        Evaluate f(θ*) using probabilistic model
        H ← H ∪ (θ*, f(θ*))
        Fit a new model M_i to H
        Prune hyperparameter search space using Hyperband
    return H
```

Fig. 3 – Pseudocode for the TPE algorithm with hyperband pruning.

$$\int_{-\infty}^{y^*}(y-y^*)p(y|\theta)p(y)dy = l(\theta)\int_{-\infty}^{y^*}(y-y^*)p(y)dy$$

$$= \gamma y^* l(\theta) - l(\theta)\int_{-\infty}^{y^*}p(y)dy \tag{15}$$

Hence, $EI_{y^*}(\theta)$ can be expressed as:

$$EI_{y^*}(\theta) = \frac{\gamma y^* l(\theta) - l(\theta)\int_{-\infty}^{y^*}p(y)dy}{\gamma l(\theta) + (1-\gamma)g(\theta)} \propto \left(\gamma + \frac{g(\theta)}{l(\theta)}(1-\gamma)\right)^{-1} \tag{16}$$

Therefore, TPE maximizes the ratio $\frac{l(\theta)}{g(\theta)}$, i.e., maximizes $l(\theta)$ and minimizes $g(\theta)$.

Pruning algorithms such as hyperband automatically stop search trials with poor predictive performance at the early stages of training. The hyperband [56] pruner iteratively selects an optimum set of hyperparameters by dynamic resource allocation, and considers a fixed computational budget (such as the number of training epochs). Hyperband operates similarly to a variable early stopping strategy that interrupts training if the model with a particular hyperparameter configuration converges to a poor solution (low accuracy) and extends training if the model performance improves (high accuracy). Since hyperband uses a random search to select the hyperparameter configurations, no information passes between trials that could be used to inform the hyperparameter selector. Instead of a random search, a Bayesian optimization technique, such as TPE, can be implemented to select the hyperparameter configuration in each trial.

Prognostics strategy

This section details the data preprocessing steps applied to the dataset, the loss function and evaluation metrics used to evaluate the prediction results, and the model development and validation workflow.

Data preprocessing

The FC1 dataset contains 143,862 data points and the FC2 dataset contains 127,370 data points. To reduce the computational complexity while processing the data and preserving the PEMFC's degradation characteristics, all measurements or data points in both datasets were sampled at 1 h. This resulted in 1155 and 1021 data points for the FC1 and FC2 datasets, respectively. The stack voltage of FC1 shows an irregular and noisy pattern while those of FC2 show significant spikes due to the dynamic loading conditions. To reduce high-frequency noise and smoothen the resulting datasets, a second-order Savitzky-Golay filter (SGF) with a window length (number of coefficients) of 21 was applied to both the FC1 and FC2 datasets. The SGF is a finite impulse response digital filter that smoothens a sequence by fitting a low-degree polynomial using a linear least-squares method [57]. The original (raw), down-sampled, and filtered stack voltages for FC1 and FC2 are shown in Fig. 4(a) and 4(b) – 4(c), respectively.

It is noteworthy that the SGF was selected over locally weighted scatterplot smoothing (LOWESS), commonly used in the PEMFC prognostics literature for noise filtering and data smoothening [58–60], as unlike the SGF, LOWESS is sensitive to outliers and requires densely sampled datasets to generate a satisfactorily denoised output [61].

Next, the failure threshold for estimating the RUL was determined. The failure threshold determines the failure point when the PEMFC stack cannot provide sufficient voltage for operation. The RUL of the PEMFC stack is thus the time for which the stack can provide an output voltage greater than or equal to its failure threshold. The failure thresholds for FC1 and FC2 were chosen as 96% of the stack voltage at 0 h, obtained after passing through the SGF. The failure threshold was selected as 96% in line with the recent literature on PEMFC prognostics [25,26,28,59]. The failure thresholds for FC1 and FC2 were computed as 3.2028 V (= 0.96 × 3.3362 V) and 3.1971 V (= 0.96 × 3.3303 V). Notably, the window length and order of the SGF affect the smoothed stack voltage and, thus, the failure threshold.

A time series dataset needs to be formulated as a supervised learning problem to use supervised deep learning models for RUL estimation. While time series contain data indexed in time, a supervised learning problem consists of input and output patterns so that a machine learning model can predict the output based on the input patterns. Creating columns of observations lagging in time by shifting each feature by one or more timesteps converts a time series dataset to a dataset suitable for supervised learning. Here, each stack measurement is lagged by one timestep. Thus, after adding a time lag of one timestep, the new dataset lengths for FC1 and FC2 become 1154 h and 1020 h, respectively.

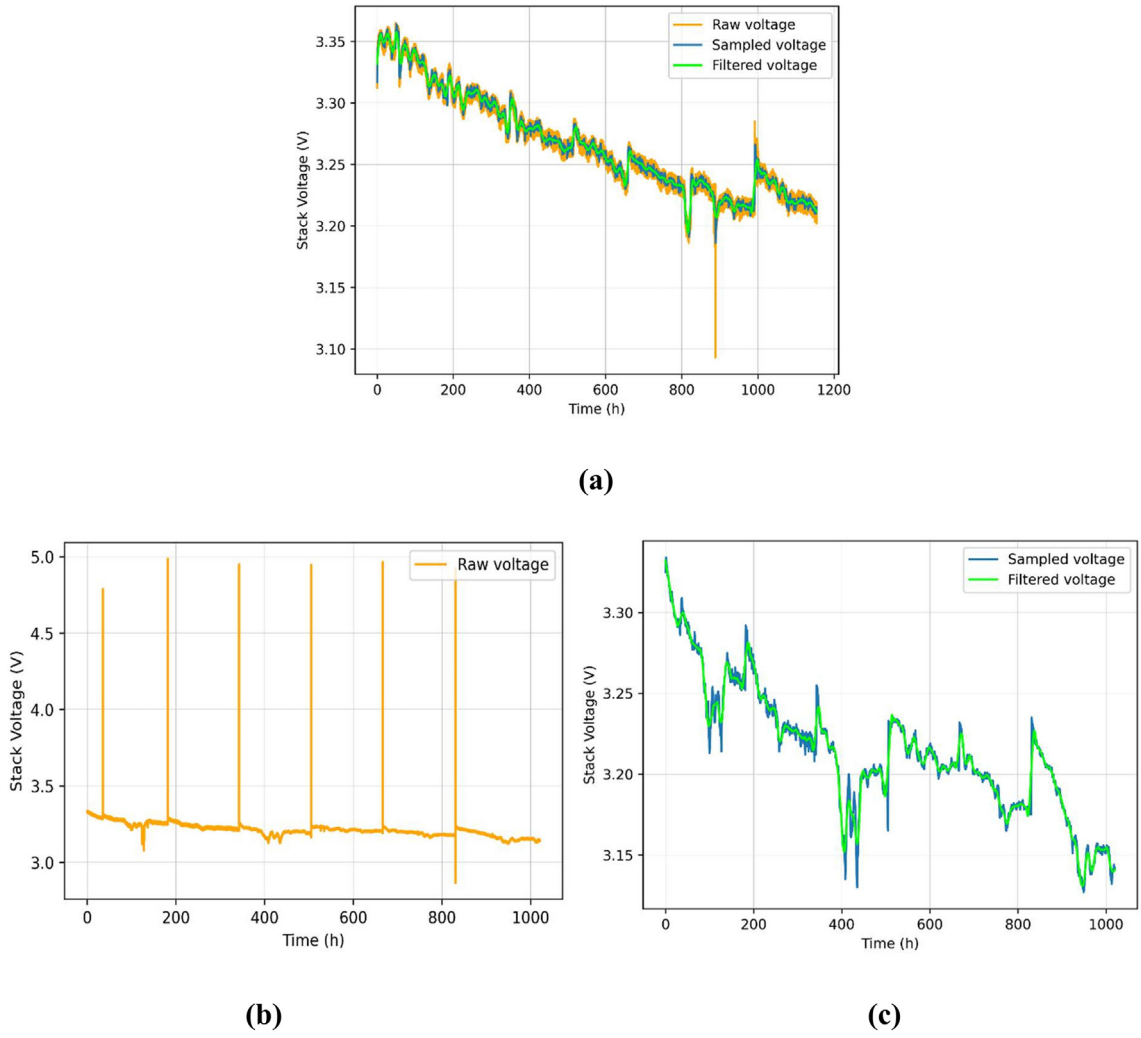


Fig. 4 – (a) Raw, sampled, and filtered voltages for FC1; (b) Raw and (c) Down-sampled and filtered voltages for FC2.

The 24 features presented in Table 1 were employed to train the developed deep learning models. As there were 24 features, including the stack voltage at the previous hour, feature selection was not employed as the number of features is relatively small, and the optimum feature importance threshold depends on the dataset statistics [62]. The FC1 and FC2 datasets were then split into training, validation, and test sets in the ratio of 50:10:40. A validation set is necessary to evaluate the model performance independent of the training and test data. The training, validation, and test set lengths (in h) for the FC1 dataset are 577 h, 115 h, and 462 h, while those for the FC2 dataset are 510 h, 102 h, and 408 h.

The validation interval is chosen so the PEMFC voltages fall below their failure threshold only after the validation duration. If the prediction starts at time $t_{predicted}$ and the actual failure or end-of-life threshold is reached at time $t_{failure}$, the actual RUL, RUL_{actual}, is given by:

$$RUL_{actual} = t_{failure} - t_{predicted} \tag{17}$$

The predicted RUL, $RUL_{predicted}$, is given by the time difference between the predicted failure threshold time $\hat{t}_{failure}$ and the prediction time:

$$RUL_{predicted} = \hat{t}_{failure} - t_{predicted} \tag{18}$$

From the smoothed datasets, the true RUL for FC1 and FC2 after the validation and training intervals were computed as 235 h and 225 h, respectively. Thus, the useful life of FC1 and FC2 was computed as 927 h and 837 h, respectively.

The training, validation, and test sets were normalized using min-max normalization. Normalization before dataset splitting can cause data leakage as statistical information, such as mean and variance, may leak into the test and validation sets from the training set. It may lead to a potential bias in the performance evaluation of the models when faced with data on the models that were not previously trained. Thus, the datasets were normalized after splitting them into training, test, and validation sets.

All features and the stack voltage in the training set were scaled to the range (0, 1) using the min-max normalization equation:

$$\overline{x} = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{19}$$

Here, x represents the original feature value and $\bar{x}$ represents the scaled feature value.

The validation and test sets were normalized using the training set's minimum and maximum values of the respective features. Although doing so may not scale the test set and validation set features to the range (0,1), this allows unbiased model performance evaluation.

Loss function and evaluation metrics

The predictive performance of each model was benchmarked using three metrics for long-term degradation prediction – the coefficient of determination (R^2), root-mean-square error (RMSE) and mean absolute percentage error (MAPE) – and the relative error (RE) in RUL estimation. Models with high predictive accuracy show high R^2 (~1) but low RMSE, MAPE, and RE. Denoting the number of voltages in the smoothed dataset as n, the original stack voltages as x_i, the predicted stack voltages by $\hat{x}_i$, and the mean of the original stack voltages as $\overline{x_i}$, the metrics R^2, RMSE, and MAPE are defined as:

$$R^2 \triangleq 1 - \frac{(x - \overline{x_i})^2}{(x - \hat{x}_i)^2} \tag{20}$$

$$RMSE \triangleq \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2} \tag{21}$$

$$MAPE \triangleq \frac{100}{n}\sum_{i=1}^{n}\left|\frac{x_i - \hat{x}_i}{x_i}\right| \% \tag{22}$$

The RE in RUL estimation are defined as:

$$RE \triangleq \frac{|RUL_{actual} - RUL_{predicted}|}{RUL_{actual}} \tag{23}$$

For hyperparameter optimization using TPE, the RMSE was chosen in the loss function, while the loss function for training each neural network model was selected as the mean absolute error (MAE), which is defined as:

$$MAE \triangleq \frac{1}{n}\sum_{i=1}^{n}|x_i - \hat{x}_i| \tag{24}$$

When the PEMFC is connected to dynamic loads, large stack voltage fluctuations may occur wherein the fluctuation may depend on the nature of the load. These fluctuations may produce RUL misjudgements, mainly if they occur near the end of the training and validation stages and/or randomly during the test stage. To avoid RUL misjudgement, the prediction results of each model were averaged over 50 training iterations rather than relying on the results of only one training iteration. Moreover, since neural networks employ randomly initialized weights, statistical averaging was necessary to extract valuable insights about the prediction accuracy of the developed models. Further, each model's initial states were reset on each training iteration so that the model does not fit the same weights and biases to the training, validation, and test sets in each iteration.

Model development and validation

Neural network implementations from the open-source Keras [63] library with a TensorFlow backend (version 2.9.2) for Python 3 were used to build the prognostic models. All models were trained on NVIDIA DGX A100 GPUs to accelerate model training. The codes for implementing the proposed models are available on GitHub at https://github.com/kar10tik/Long-Term-PEMFC-DL-Prognostics. Overfitting during training was prevented by dropout regularization and using an early stopping training strategy. Overfitting occurs when a model shows excellent prediction accuracy on the training set but poor accuracy on the test set as it memorizes the training set data rather than fitting to it. Dropout randomly drops neurons and their connections during training to prevent overfitting. In early stopping, training is stopped when the validation loss of the models stops improving for a fixed number of epochs known as the patience value. The patience value specifies the number of epochs for which no improvement in the validation loss is observed, after which the model stops training. The dropout was selected as a hyperparameter to be optimized and each model was trained for 150 epochs with early stopping with a patience value of 30.

For fast convergence of prediction results, three gradient-based neural network optimizers were explored for their prediction performance – stochastic gradient descent (SGD), Root Mean Squared propagation (RMSprop), and Adam. SGD [64] is a gradient-based optimizer that computes the gradient of the objective function from a stochastic section of the data rather than the entire dataset. The algorithm follows the negative gradient of an objective function to locate its global minimum. The RMSprop optimizer [65] uses a decaying average of partial gradients to modify the step size for each parameter so that the optimizer forgets early gradients to focus on the most recently observed partial gradients. The adaptive moment optimizer, also known as Adam optimizer [66], is a variant of RMSprop that uses a variable learning rate adjusted based on the exponentially weighted average of the gradients. However, the best optimizer for a neural network model depends on the network architecture and the dataset used to train the network.

The implementation workflow for the proposed models is depicted in Fig. 5. The open-source implementation of TPE with hyperband pruning in Optuna [67] for Python was used to optimize the prognostic models. Each model was trained over 500 trials with an early stopping strategy in each model with a patience value of 30. For each model, the dropout was searched in the range of [0.01, 0.5] while the optimizer was searched in the set {SGD, Adam, RMSProp}. The learning rate for the SGD and Adam optimizers was searched in the logarithmically spaced range of [1E-5, 1E-1]. For the SGD optimizer, the momentum was searched in the logarithmically spaced range of [1E-3, 99E-2]. For the RMSprop optimizer, the decay parameter was searched in the logarithmically spaced range of [0.85, 0.99].

For the LSTM and GRU models, the optimal number of hidden layers (i.e., LSTM or GRU layers) was searched in the range of [1, 10]. The number of neurons or units in each hidden

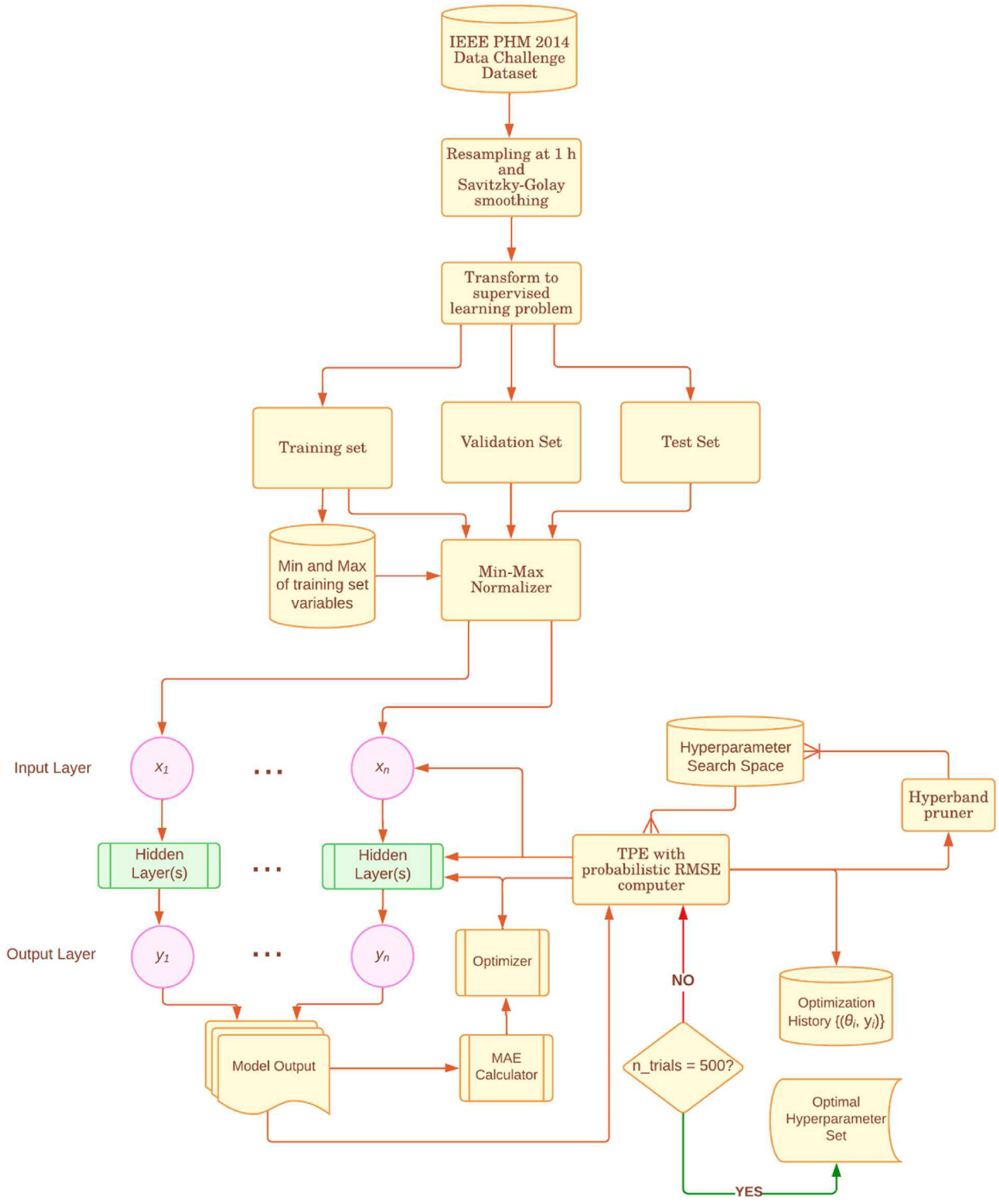


Fig. 5 – Implementation of the proposed deep learning-based prognostic approach.

layer was searched in the range of [30, 100] and each hidden layer's activation function was set to tanh. For the 1D–CNN–LSTM, 1D–CNN–GRU, 1D–CNN–Bidirectional LSTM, and 1D–CNN–Bidirectional GRU models, the activation function for the convolutional layer was searched in the set {'tanh', 'softmax', 'relu'} and the number of convolution filters was searched in the range [30, 100]. For these models, the number of neurons was searched in the range [30, 110].

In summary, the model development and validation workflow is as follows.

1. The FC1 and FC2 datasets were down-sampled at an interval of 1 h and then smoothed and denoised using a second-order SGF with a window length of 21.
2. Both datasets were transformed into a supervised learning problem by lagging the measurements by one timestep. It resulted in 24 features (stack measurements at time $t-1$) and one dependent variable to be predicted (total stack voltage at time t).
3. Both datasets were split into training, validation, and test sets in the ratio 50:10:40 and then normalized using min-max normalization with the minimum and maximum of each feature selected from their respective training sets.
4. Two sets of six models – one set for FC1 and FC2 each – were developed with the normalized training and validation datasets and the MAE as the loss function.
5. The hyperparameter search space was defined as described above. A set of 500 trials was run to optimize the hyperparameters of each model with RMSE in the degradation trend prediction as the loss function. In each trial, the models were trained for 150 epochs with early stopping with a patience value of 30.
6. For FC1 and FC2, the models with optimized hyperparameters were tested on the test data and the prediction results were averaged over 50 trials.
7. The optimized models for FC1 were tested for their prediction performance on the FC2 dataset and the prediction results were averaged over 50 trials.

Prediction results and discussion

This section presents the results of hyperparameter optimization, the predictive performance of each model on the FC1 and FC2 datasets, and the transfer learning performance of models trained on FC1.

Hyperparameter optimization results

The hyperparameter configurations, found by TPE with hyperband pruning over 500 search trials on the defined hyperparameter search space, are listed for the FC1 and FC2 datasets in Tables 2 and 3, respectively. The optimum number of hidden layers for the LSTM and GRU models was interestingly computed as 1 for both FC1 and FC2 datasets. For all evaluated models, the maximum number of hidden layers, including convolutional layers, was computed as 2. For the LSTM and GRU models, increasing the number of hidden layers to more than two resulted in high computational complexity and poor predictive performance. As each layer's units were also considered as a hyperparameter, the hyperparameter search space increased with the number of hidden layers. Interestingly, for all models, the best-fitting activation function and the best-fitting optimizer for both FC1 and FC2 datasets were found to be tanh and Adam, respectively.

For models trained on the FC1 dataset, the optimal dropout ranged from 0.06 to 0.31, while for models trained on the FC2 dataset, the optimal dropout ranged from 0.01 to 0.09. The optimal learning rate for models trained on the FC1 dataset ranged from 1.20E-4 to 6.90E-4, while the optimal learning rate for models trained on the FC2 dataset ranged from 1.80E-4 to 4.05E-4.

The average time to compute the best hyperparameter configuration for all evaluated models was 2 h. However, most models converged RMSEs close to those of their best hyperparameter configurations within 1.25 h when nearly 300 trials were complete. Therefore, around 300 trials may be used for online model optimization after around half of the degradation data is obtained from the PEMFC system.

Table 2 – Optimum hyperparameter values for models trained on the FC1 dataset.

Model architecture	Hidden layers	Optimizer learning rate	Dropout	Units in recurrent hidden layer	Filters for CNN layer
LSTM	1	4.5967E-4	10.204E-2	75	–
GRU	1	6.8995E-4	6.0171E-2	92	–
1D–CNN–LSTM	2	3.9458E-4	21.587E-2	108	93
1D–CNN–GRU	2	5.6239E-4	6.864E-2	102	97
1D–CNN–B-LSTM[a]	2	1.4155E-4	31.165E-2	100	79
1D–CNN–B-GRU[a]	2	1.1281E-4	22.507E-2	71	95

[a] B: Bidirectional.

Table 3 – Optimum hyperparameter values for models trained on the FC2 dataset.

Model architecture	Hidden layers	Optimizer learning rate	Dropout	Units in recurrent hidden layer	Filters for CNN layer
LSTM	1	3.0787E-4	2.4500E-2	82	–
GRU	1	3.3981E-4	3.0337E-2	68	–
1D–CNN–LSTM	2	4.0525E-4	8.7344E-2	104	92
1D–CNN–GRU	2	1.8069E-4	1.0176E-2	107	88
1D–CNN–B-LSTM[a]	2	1.7985E-4	4.3454E-2	97	91
1D–CNN–B-GRU[a]	2	1.8921E-4	1.0854E-2	107	100

[a] B: Bidirectional.

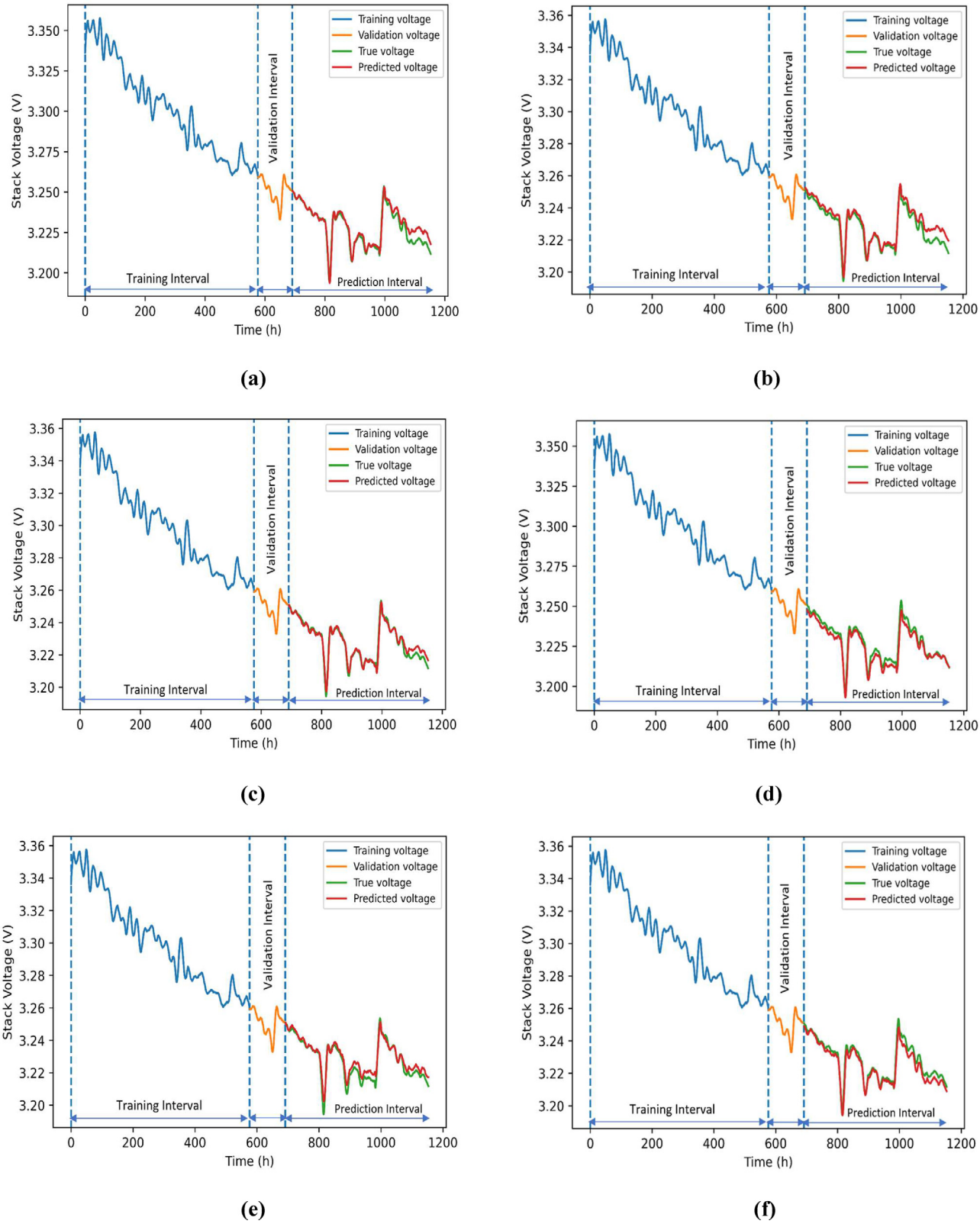


Fig. 6 – Predicted stack voltage curves for FC1 by (a) LSTM; (b) GRU; (c) 1D–CNN–LSTM; (d) 1D–CNN–GRU; (e) 1D–CNN–Bidirectional-LSTM; (f) 1D–CNN–Bidirectional-GRU.

Table 4 – Prognostic results for FC1 dataset iterated over 50 trials (true RUL = 235 h).

Model architecture	RMSE (× E−3)			MAPE (× E−1)			R^2 (× E−2)			Predicted RUL (h)			Relative error in RUL (× E−1)		
	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean
LSTM	1.258	7.307	2.815	0.285	1.744	0.759	61.99	98.87	93.42	236	577	250.28	0.042	14.55	0.650
GRU	1.494	7.102	2.815	0.332	1.290	0.720	64.09	98.41	93.14	234	238	236.26	0.000	0.128	0.055
1D−CNN−LSTM	1.606	5.001	2.896	0.372	1.196	0.623	77.43	98.16	93.50	235	238	236.70	0.000	0.128	0.072
1D−CNN−GRU	1.564	7.579	3.480	0.420	2.061	0.899	59.10	98.26	89.66	233	577	249.74	0.000	14.55	0.636
1D−CNN−B-LSTM[a]	1.632	6.872	3.579	0.390	2.779	0.889	66.37	98.10	89.88	236	577	339.42	0.042	14.55	4.443
1D−CNN−B-GRU[a]	1.624	9.356	3.133	0.323	0.754	1.675	37.67	98.12	91.70	235	238	236.54	0.000	0.127	0.065

[a] B: Bidirectional.

Prediction results for FC1

The training, validation, and test intervals for models trained on FC1 are 577 h, 115 h, and 462 h, respectively. The prediction results shown in Fig. 6 demonstrate that the predicted voltage curves for each of the six investigated models trained on FC1 accurately follow the variation of the stack voltage, offering important insights into the stack voltage degradation trend beforehand. Table 4 shows the RMSE, MAPE, R^2, predicted RUL, and RE in RUL prediction for the test dataset.

For FC1, the LSTM and GRU models revealed the best average RMSE of 2.815E-3. The GRU model also exhibited the best average RUL estimation RE of 0.055E-1. The 1D−CNN−LSTM model showed the best average R^2 and MAPE of 93.50E-2 and 0.623E-1, respectively. The best minimum RMSE of 1.258E-3, MAPE of 0.285E-1, and maximum R^2 of 98.87 were demonstrated by the LSTM model. The best minimum

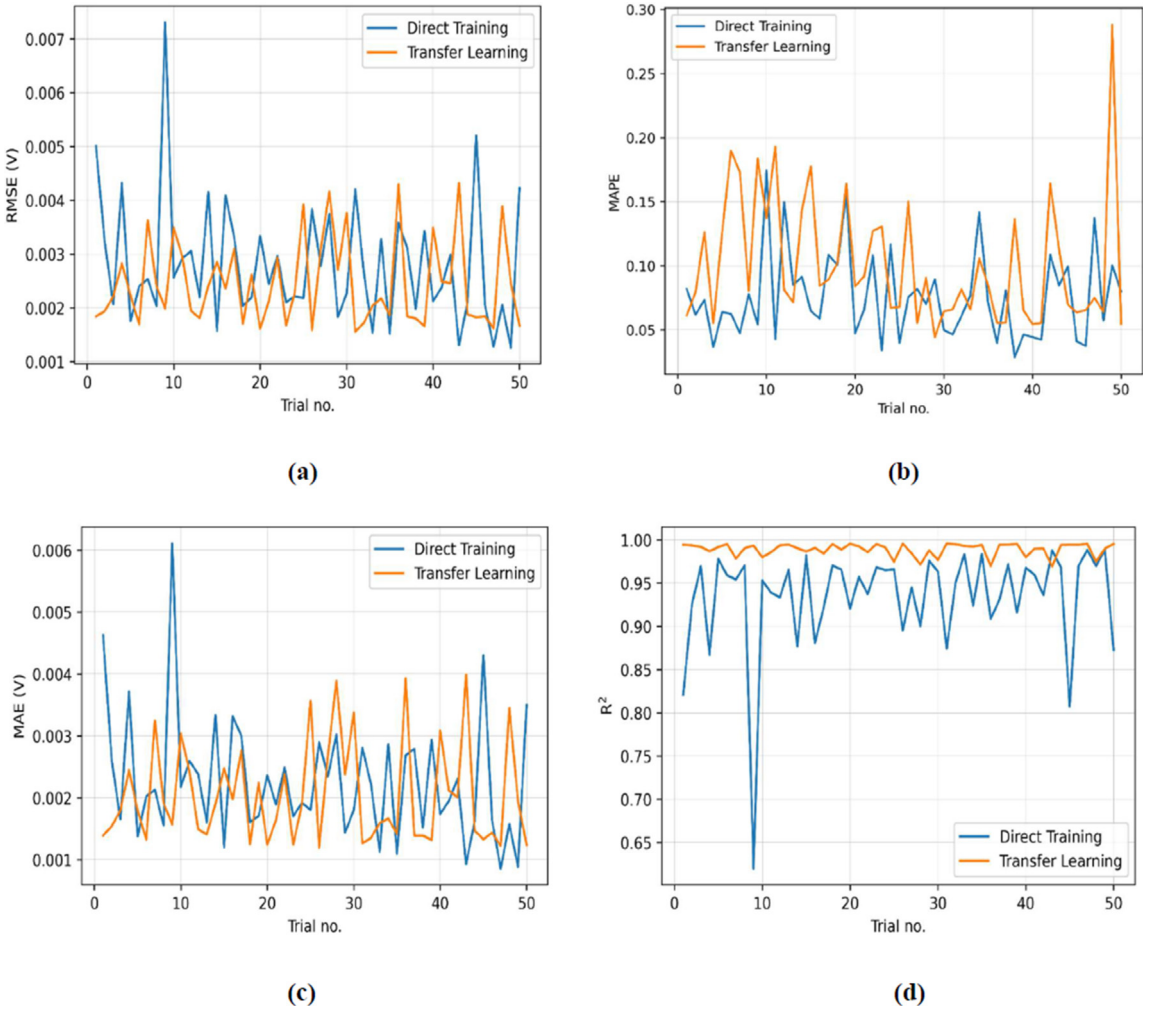


Fig. 7 – (a) RMSE (V), (b) MAPE, (c) MAE (V), and (d) R^2 in stack voltage degradation prediction for direct training on FC1 and transfer learning on FC2 of the LSTM model.

RUL prediction RE of 0.000 was exhibited by the GRU, 1D–CNN–LSTM, 1D–CNN–GRU, and 1D–CNN–Bidirectional-GRU models. The maximum predicted RUL of 577 h by the LSTM, 1D–CNN–GRU, and 1D–CNN–Bidirectional-LSTM models implies that the predicted voltages in one or more trials did not fall below the 3.2028 V (96%) threshold throughout the prediction interval. Also, the mean RMSE increased upon adding a CNN layer to the LSTM and GRU models. However, the mean MAPE of the 1D-CNN-LSTM model was lower than that of the LSTM model, while that of the 1D-CNN-GRU model was higher than that of the GRU model. The 1D–CNN–Bidirectional-LSTM and 1D–CNN–Bidirectional-GRU models also show higher mean RMSE and MAPE and lower mean R^2 than the LSTM and GRU models. However, the 1D–CNN–LSTM model showed a lower RUL prediction RE than the LSTM model, while the 1D–CNN–GRU model showed a higher RUL

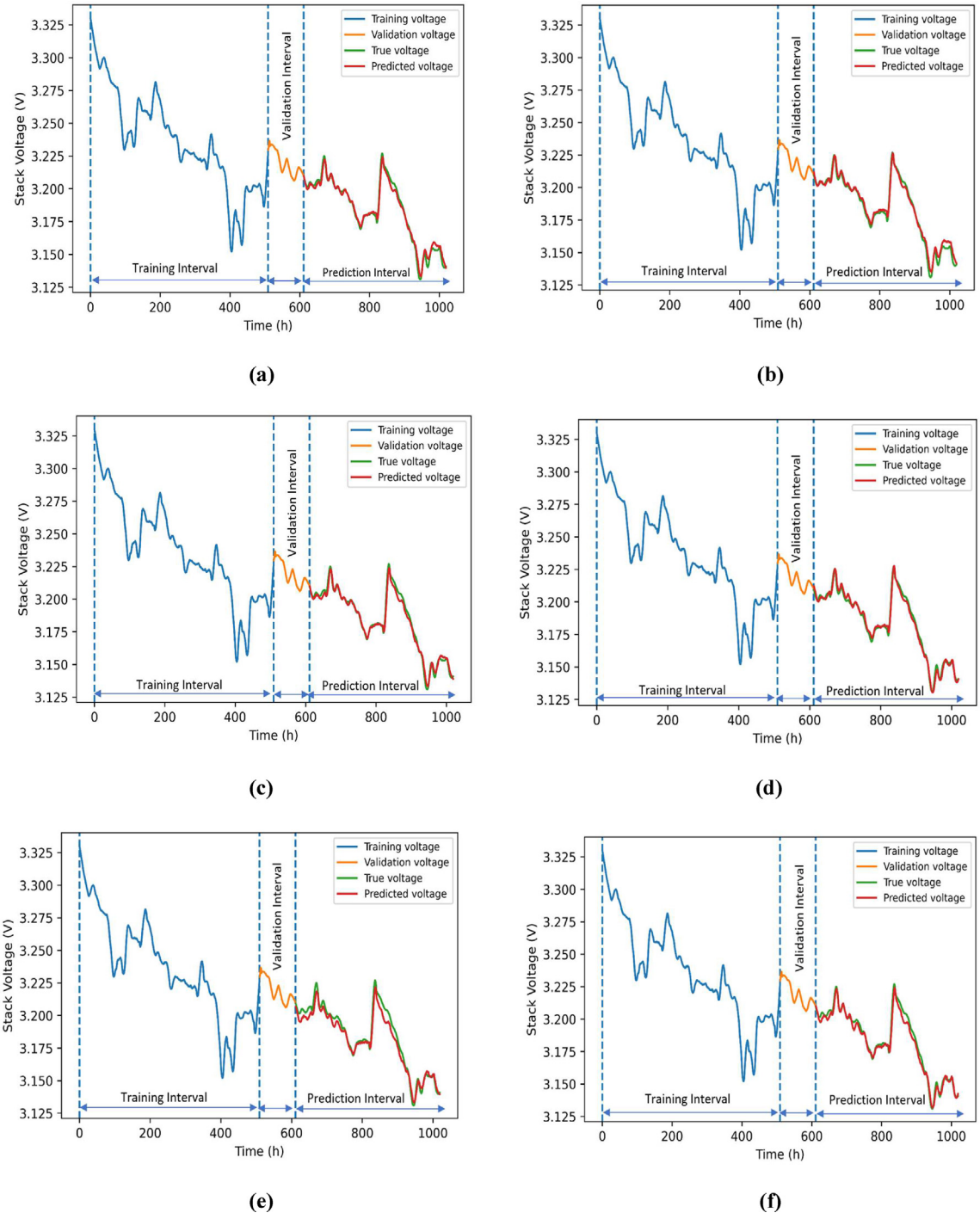


Fig. 8 – Predicted stack voltage curves for FC2 by (a) LSTM; (b) GRU; (c) 1D–CNN–LSTM; (d) 1D–CNN–GRU; (e) 1D–CNN–Bidirectional-LSTM; (f) 1D–CNN–Bidirectional-GRU.

Table 5 – Prognostic results for FC2 dataset iterated over 50 trials (true RUL = 225 h).

Model architecture	RMSE (× E−3)			MAPE (× E−1)			R^2 (× E−2)			Predicted RUL (h)			Relative error in RUL (× E−1)		
	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean
LSTM	2.331	8.102	3.352	0.552	2.491	0.837	89.22	99.11	97.99	110	229	197.60	0.044	5.200	1.235
GRU	1.812	12.84	3.348	0.480	4.615	0.986	72.92	99.46	97.73	104	228	175.94	0.044	5.378	2.204
1D–CNN–LSTM	2.445	5.816	4.041	0.568	1.869	1.064	94.45	99.02	97.18	108	228	140.84	0.044	5.200	3.749
1D–CNN–GRU	1.901	9.172	3.812	0.472	1.984	1.046	86.19	99.41	97.26	107	230	165.72	0.044	5.244	2.658
1D–CNN–B-LSTM[a]	1.956	7.017	3.351	0.423	2.663	1.116	91.92	99.37	97.98	108	234	189.02	0.133	5.200	1.626
1D–CNN–B-GRU[a]	1.776	8.807	3.398	0.444	2.882	1.015	87.27	99.48	97.68	107	229	170.96	0.089	5.245	2.441

[a] B: Bidirectional.

prediction RE than the GRU model. The mean predicted RUL closest to the actual RUL (235 h) was demonstrated by the GRU model (236.26 h). The RMSE in volt (V), MAPE, MAE (V), and R^2 for direct training on FC1 and transfer learning on FC2 of the LSTM model have been shown in Fig. 7.

Prediction results for FC2

The training, validation, and test intervals for models trained on FC2 are 510 h, 102 h, and 408 h, respectively. The prediction results shown in Fig. 8 suggest that the predicted voltage

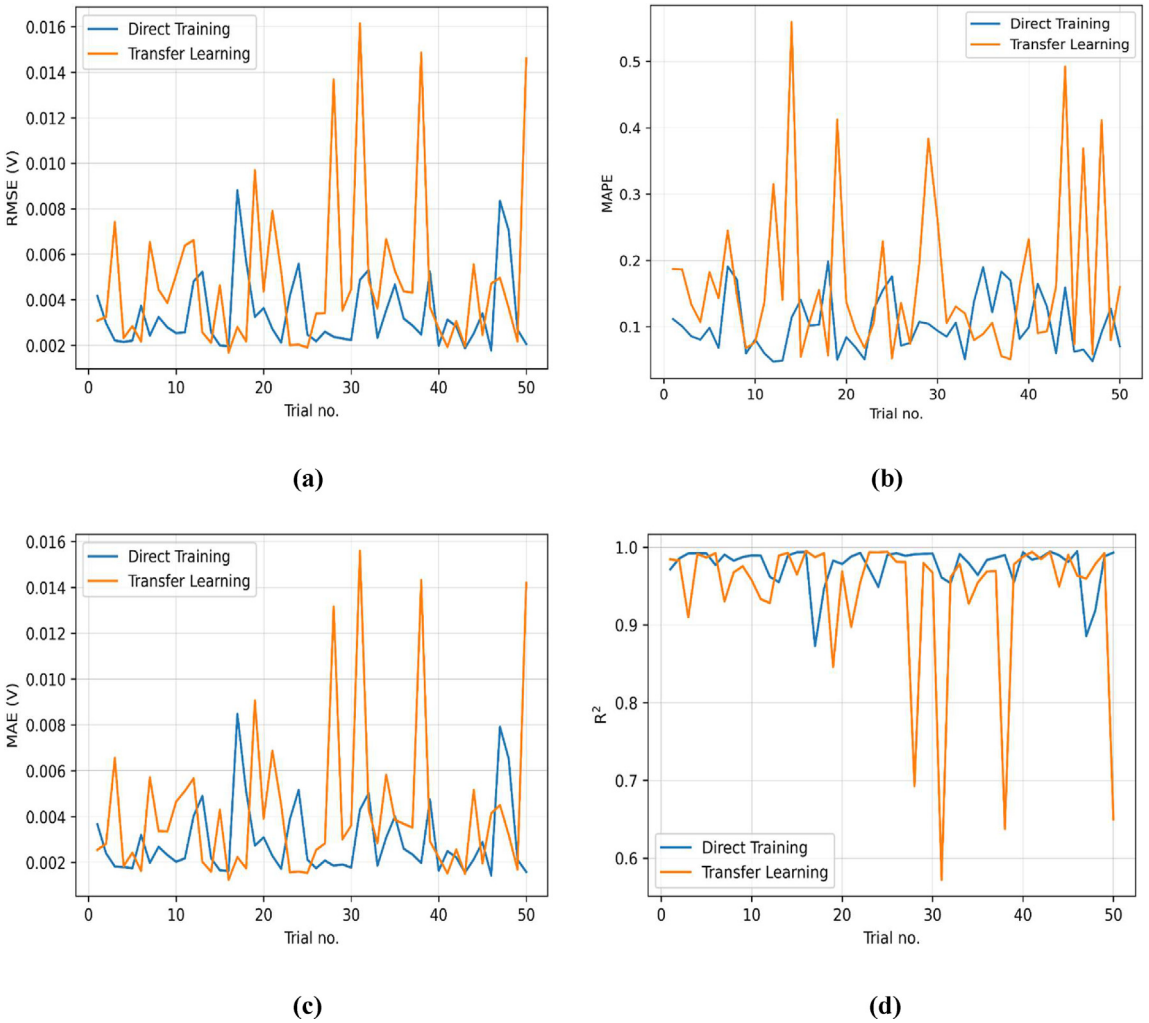


Fig. 9 – (a) RMSE (V), (b) MAPE, (c) MAE (V), and (d) R^2 in stack voltage prediction of the 1D–CNN–Bidirectional-GRU model for direct training on FC2 and transfer learning on FC2 after training on FC1.

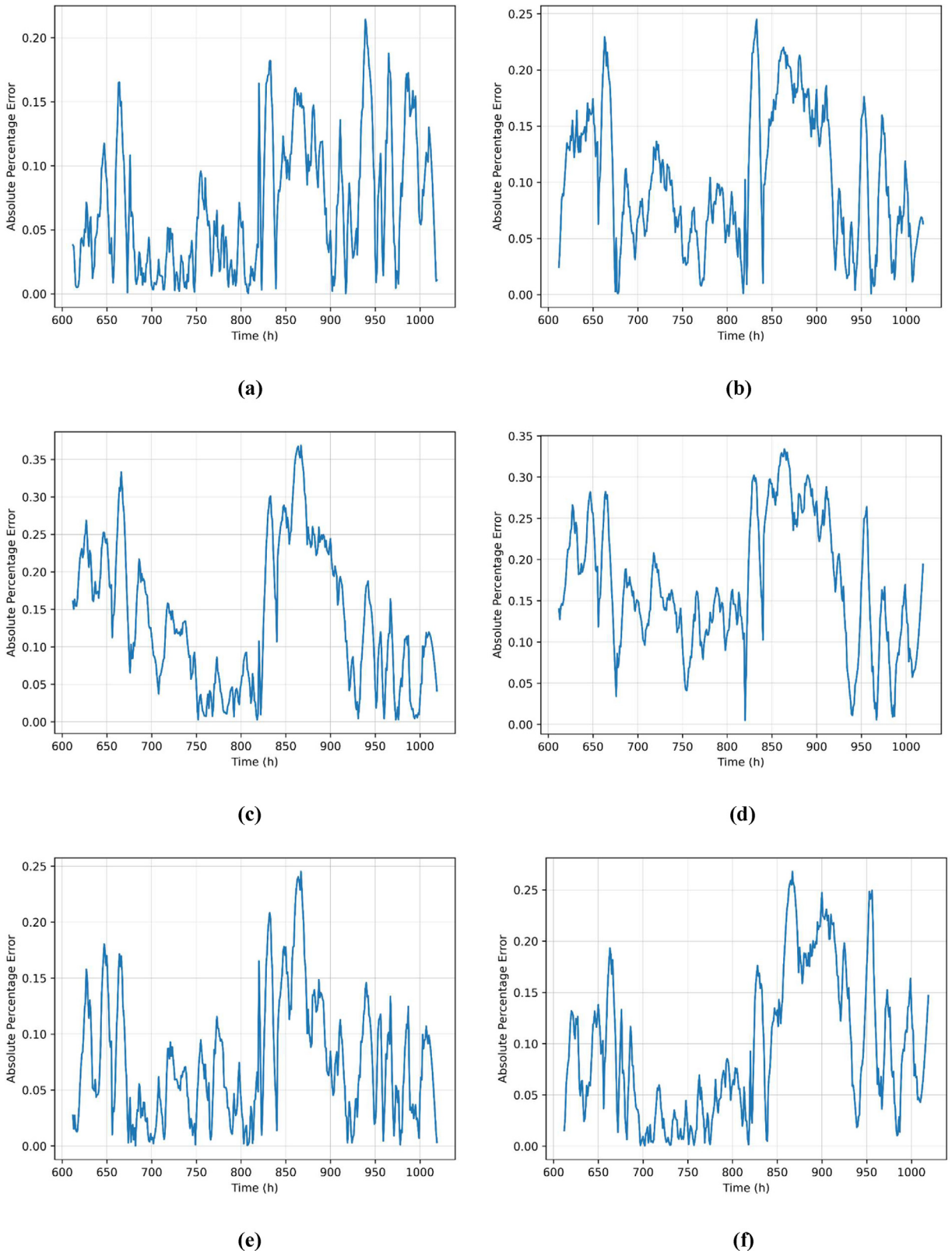


Fig. 10 – Absolute percentage error in the predicted stack voltage versus prediction time for direct training on FC2 of (a) LSTM; (b) GRU; (c) 1D–CNN–LSTM; (d) 1D–CNN–GRU; (e) 1D–CNN–Bidirectional-LSTM; (f) 1D–CNN–Bidirectional-GRU.

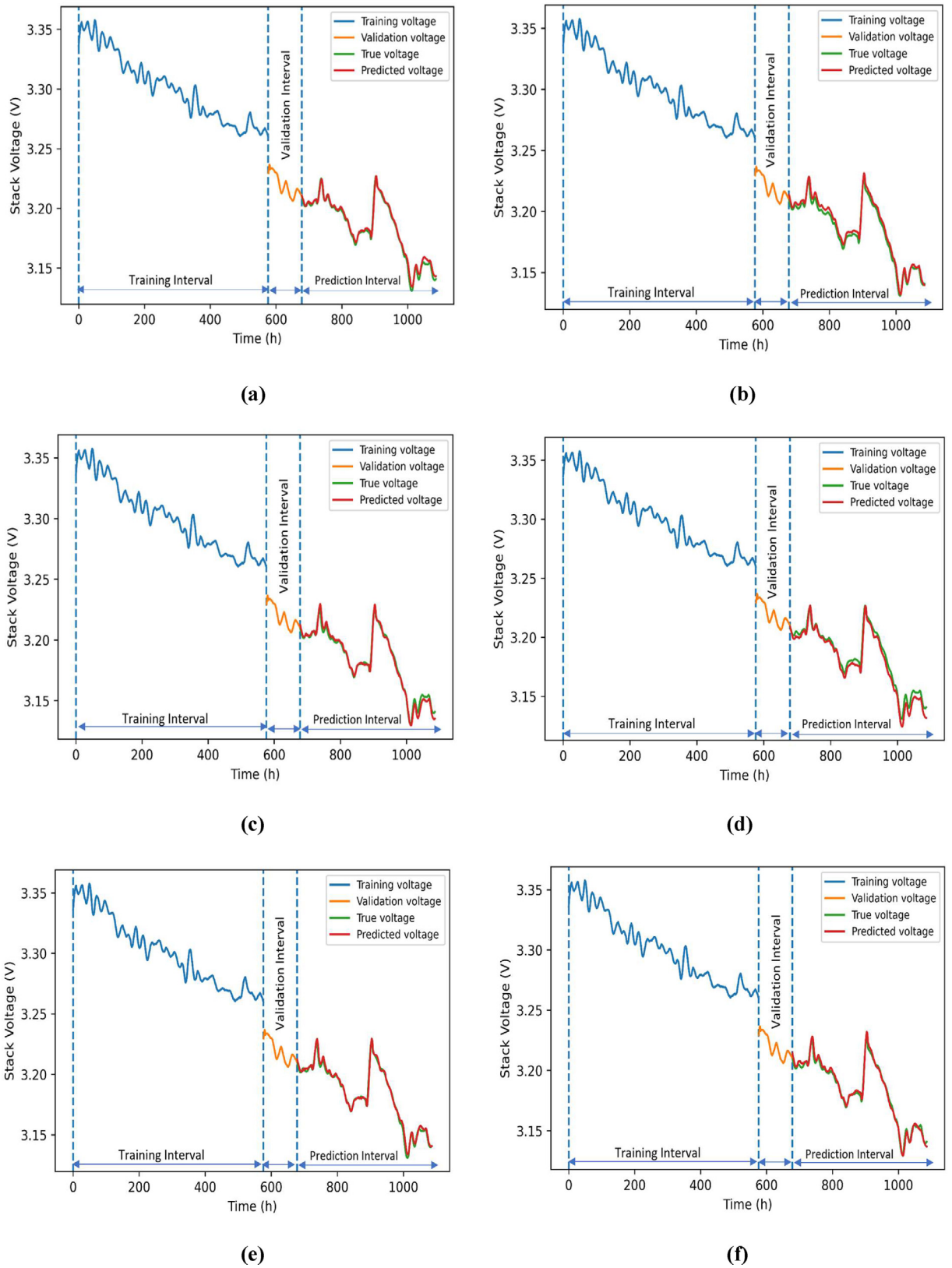


Fig. 11 – Predicted stack voltage curves by transfer learning of (a) LSTM; (b) GRU; (c) 1D–CNN–LSTM; (d) 1D–CNN–GRU; (e) 1D–CNN–Bidirectional-LSTM; (f) 1D–CNN–Bidirectional-GRU.

Table 6 – Transfer learning prognostic results iterated over 50 trials (true RUL = 225 h).

Model architecture	RMSE (× E−3)			MAPE (× E−1)			R^2 (× E−2)			Predicted RUL (h)			Relative error in RUL (× E−1)		
	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean
LSTM	1.555	4.317	2.455	0.362	1.215	0.630	96.94	99.60	98.91	202	229	210.28	0.044	1.022	0.697
GRU	1.777	7.638	3.495	0.346	2.537	0.984	90.42	99.48	97.67	109	242	202.78	0.000	5.155	1.172
1D−CNN−LSTM	1.850	8.433	4.175	0.482	2.966	1.172	88.33	99.44	96.76	110	239	198.04	0.000	5.111	1.342
1D−CNN−GRU	2.077	10.61	4.901	0.446	3.771	1.441	81.51	99.29	95.26	107	242	201.22	0.044	5.244	1.393
1D−CNN−B-LSTM[a]	1.917	12.54	3.351	0.476	4.493	1.312	74.19	99.40	95.91	107	241	204.54	0.000	5.244	1.162
1D−CNN−B-GRU[a]	1.686	16.14	4.864	0.510	5.593	1.654	57.22	99.53	94.20	102	241	198.12	0.000	5.467	1.394

[a] B: Bidirectional.

curves of each of the six investigated models trained on FC2 accurately follow the stack voltage variation. The RMSE, MAPE, R^2, predicted RUL, and RE in RUL prediction for the test dataset are presented in Table 5.

Of all models trained on FC2, the GRU model exhibited the best average RMSE of 3.348E-3, while the LSTM model exhibited the best average MAPE, RUL estimation RE, and R^2 of 0.837E-1, 1.235E-1, and 97.99E-2, respectively. The average R^2 of 97.98E-2 shown by the 1D-CNN-Bidirectional-LSTM model is comparable to that of the LSTM model. The RMSE (V), MAPE, MAE (V), and R^2 for direct training on FC2 and transfer learning on FC2 after training on FC1 of the 1D−CNN−Bidirectional-GRU model have been shown in Fig. 9. The 1D−CNN−Bidirectional-GRU model showed the best minimum RMSE of 1.776E-3 and maximum R^2 of 99.48E-2. The 1D−CNN−Bidirectional-LSTM model showed the best minimum MAPE of 0.423E-1. The best minimum RUL prediction RE of 0.044E-1 was demonstrated by the LSTM, GRU, 1D−CNN−LSTM, and 1D−CNN−GRU models.

Adding a CNN layer to the LSTM and GRU models increased the mean RMSE, MAPE, and RUL prediction RE but decreased the mean R^2 compared to the LSTM and GRU models, respectively. Hence, the 1D−CNN−LSTM and 1D−CNN−GRU models showed lower transfer learning prediction accuracy than the LSTM and GRU models. The average predicted RUL closest to the true RUL of 225 h−197.60 h – was demonstrated by the LSTM model. Since the model with the least minimum RMSE was chosen as the best-performing model, the 1D−CNN−Bidirectional-GRU model exhibited the best overall predictive performance for FC2.

Fig. 10 depicts the absolute percentage test-phase prediction error for direct training on FC2 for each of the proposed models as a function of the test-phase time. Since the prediction errors during the training and validation phases converged to zero at the end of the respective phases for all models, the training and validation prediction errors are not depicted. These plots are obtained for one training-validation-test iteration for each model and may vary with each iteration due to the stochastic weights and biases of neural networks. From the plots, it may be inferred that for all models, sharp peaks in the error plots appear around 850 h due to a sudden peak in the stack output voltage in Fig. 4(c). However, the absolute percentage error does not exceed 0.40 for any model. This clearly indicates that the proposed prognostics models demonstrate robust predictive performance in direct training.

Transfer learning results

In sections 4.2 and 4.3, the prediction results of the two sets of six models each – one set for FC1 and the other for FC2 – were analyzed for their long-term degradation trend and RUL predictive performance. To save computational resources and increase generalization ability, deep learning models may be trained on one dataset and used to predict observations from another dataset via transfer learning. Each deep learning model trained on FC1 was tested for its long-term degradation and RUL prediction results on FC2 to analyze the generalizability of the models developed under static loading to dynamic loading. The six models trained on FC1 using the optimum hyperparameters listed in Table 2 were validated and tested on FC2 validation and test sets. The degradation trend and RUL prediction results of the six models are shown in Fig. 11 and Table 6. The voltage curves of Fig. 11 are discontinuous as the training set is from the FC1 dataset, while the validation and test sets are from the FC2 dataset. The loss metrics in Table 6 show that the models developed by transfer learning are more accurate and reliable than the ones in section 4.2. However, the prediction accuracy of transfer learning models, and hence the generalizability of the developed models, depends on the similarity between the training and test datasets. The FC1 and FC2 datasets are derived from PEMFC stacks with identical characteristics subjected to different loading conditions. Moreover, the two stacks follow a similar voltage degradation trend but with varying peaks at different times.

The LSTM model demonstrated the best minimum and average RMSE of 1.555E-3 and 2.455E-3, while the GRU model exhibited the best minimum MAPE of 0.346E-1. The LSTM model also exhibited the best average MAPE and R^2 of 0.630E-1 and 98.91E-2, respectively. The least RE in RUL estimates of 0.0 was exhibited by the GRU, 1D−CNN−LSTM, 1D−CNN−Bidirectional-LSTM, and 1D−CNN−Bidirectional-GRU models. However, the best average RE in RUL of 0.697E-1 and RUL closest to the true RUL of 225 h−210.18 h – were exhibited by the LSTM model. Adding a CNN layer increased the average RMSE for the LSTM model but decreased the average RMSE for the GRU model. From these results, it may be concluded that the LSTM model showed the best overall transfer learning performance.

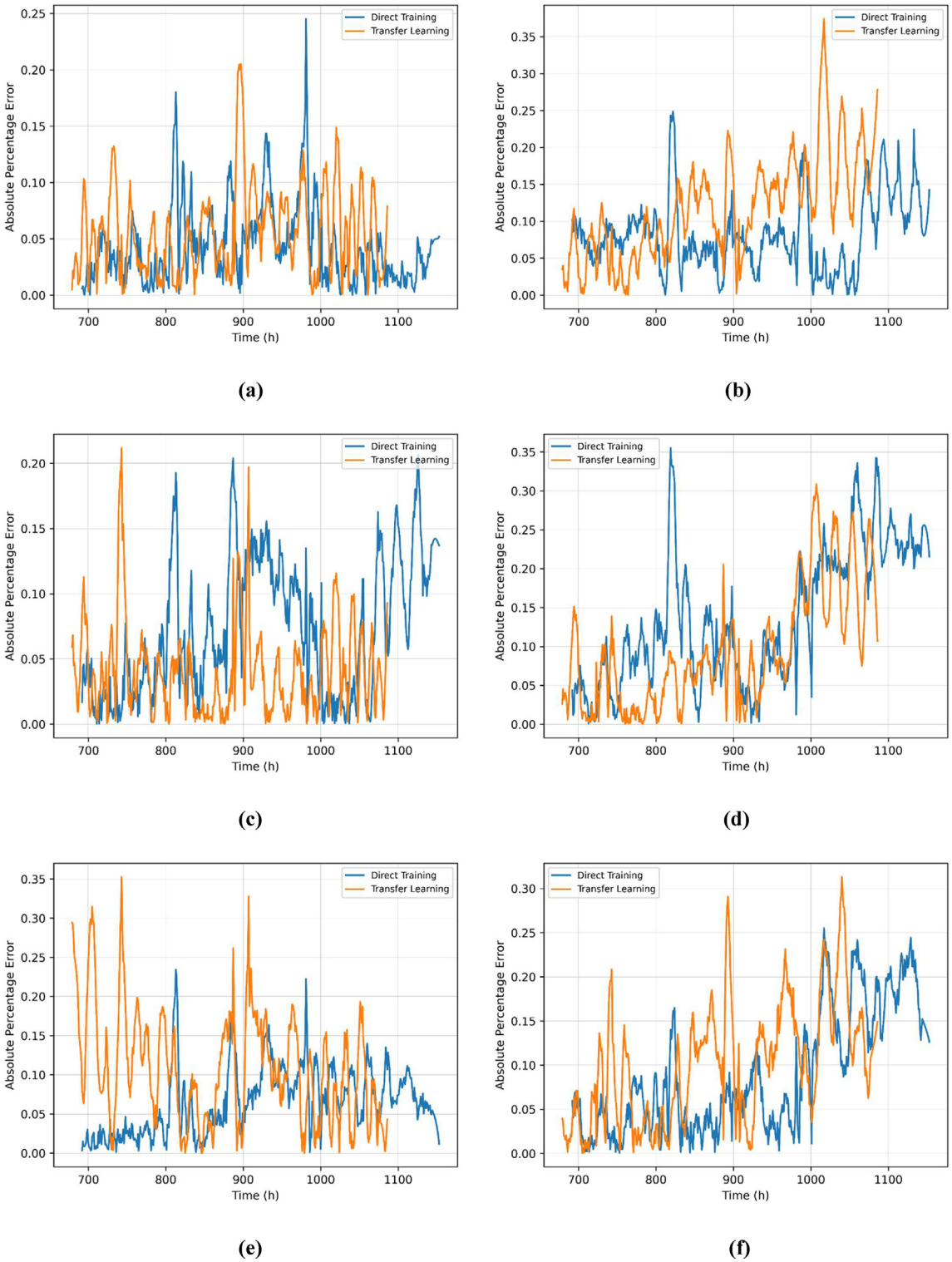


Fig. 12 – Absolute percentage error in the predicted stack voltage versus prediction time for direct training on FC1 and transfer learning on FC2 of (a) LSTM; (b) GRU; (c) 1D–CNN–LSTM; (d) 1D–CNN–GRU; (e) 1D–CNN–Bidirectional-LSTM; (f) 1D–CNN–Bidirectional-GRU.

Table 7 – Comparison of average prediction results of the proposed models with recent works.

Prognostics approach	Model architecture	Dataset	Train-validation-test split (h)	Best RMSE (× E−3)	Best MAPE (× E−1)	Best RUL prediction RE (× E−1)
Ref. [24]	B-LSTM with attention[a]	FC1	692-0-462	2.300	0.340	1.900
		FC2	721-0-299	3.400	0.533	0.900
Ref. [25]	B-LSTM-GRU with ESN[a]	FC1	200-0-954	23.50	N/R[b]	0.120
		FC2	500-0-520	11.70	N/R[b]	N/A
Ref. [26]	AT-LSTM-SVR-RF	FC1	592-0-400	4.900	1.006	0.125
		FC2	620-0-400	5.300	0.903	0.067
Ref. [28]	Dilated CNN with attention	FC1	591-0-563	4.400	0.702	7.940
	(5-step ahead prediction)	FC2	681-0-339	5.500	0.973	1.366
Proposed Model	LSTM	FC1	577-115-462	1.258	0.285	0.042
	1D–CNN–B-GRU[a]	FC2	510-102-408	1.776	0.444	0.089

[a] B: Bidirectional.
[b] N/R: Not Reported.

Fig. 12 presents the absolute percentage test-phase prediction error for direct training on FC1 and transfer learning on FC2 for each proposed model as a function of the test-phase time. As mentioned for FC2, the training and validation prediction errors are not presented and the plots represent the errors for one training-validation-test iteration. From the plots, it can be inferred that the transfer learning error plots lag the direct training error plots as the prediction timeframe for FC1 is from 692 h to 1154 h while that for FC2 is from 612 h to 1020 h. It is observed that the transfer learning and direct training absolute percentage errors did not exceed 0.40 for any model. Another observation is that the transfer learning absolute percentage error is mostly higher than the direct training absolute percentage error for the 1D–CNN–Bidirectional-LSTM and 1D–CNN–Bidirectional-GRU models. Overall, the error plots demonstrate that all models are robust in direct training and transfer learning performance.

Comparison with recent works

The proposed prediction models with the best prediction results for FC1 and FC2 – LSTM and 1D–CNN–Bidirectional-GRU, respectively – are compared with four recent works on deep learning-based prognostics employing the IEEE PHM 2014 Data Challenge Datasets for model development and validation. Notably, several works did not report whether their reported values of the evaluation metrics are the average, median, or minimum values over a certain number of training iterations. Also, most models did not report using a validation set independent of the test set for model training. Moreover, only a few works reported whether several independent training iterations were conducted to obtain the minimum, maximum, and average prediction results. Also, the actual RUL differs with variations in data preprocessing methods and hence only the RUL estimation RE was used for comparison. Therefore, the best (minimum) RMSE, MAPE, and RE in RUL prediction of the two best-proposed models – LSTM for FC1 and 1D–CNN–Bidirectional-GRU for FC2 – are compared with the respective metrics as reported in the literature.

Table 7 presents the comparison results of the proposed model with the following works – bidirectional LSTM with attention [24], bidirectional LSTM-GRU with ESN [25], LSTM-SVR-RF ensemble [26], and dilated CNN [28]. Wang et al. [24] investigated LSTMs, LSTMs with attention, and bidirectional LSTMs with attention mechanisms for degradation and RUL prediction of FC1 and FC2 after random forest feature selection with a threshold of 6.0. Two sets of training split points – 591 h and 691 h for FC1 and 620 h and 720 h for FC2 – were investigated for degradation prediction, while two other sets of training split points – 592 h and 692 h for FC1 and 621 h and 721 h for FC2 – were used for RUL estimation. The degradation prediction results showed that the bidirectional LSTM model with attention trained on 692 h of FC1 showed the lowest RMSE and MAPE of 2.300E-3 and 0.340E-1, respectively, while the LSTM model with attention trained on 721 h of FC2 showed the lowest RMSE and MAPE of 3.400E-3 and 0.533E-1, respectively. The bidirectional LSTM model with attention showed the best RUL estimation RE of 1.9E-1 and 0.9E-1 trained on 592 h of FC1 and 692 h of FC2, respectively. Li et al. [25] proposed a bidirectional-LSTM-GRU model with an ESN for degradation prediction and RUL estimation of FC1 and FC2 using only the first 200 h of training data. Three other training split points – 300 h, 400 h, and 500 h – were investigated on the datasets smoothed by a third-order SGF. The bidirectional-LSTM-GRU-ESN model, with its hyperparameters optimized using PSO and trained on the first 200 h of FC1, exhibited the best RMSE, MAE, and R^2 of 23.50E-3, 19.50E-3, and 98.22E-2, respectively. However, the authors did not report the MAPE in the predicted stack voltage.

The AT-LSTM-SVR-RF ensemble model proposed by Wang et al. [26] exhibited better RUL and degradation prediction performance than the individual models – stacked LSTM with attention, support vector regression and random forest – with the best RMSE, MAPE, and RUL prediction RE of 4.900E-3, 1.006E-1, and 0.125E-1, respectively, for FC1 and 5.300E-3, 0.903E-1, and 0.067E-1, respectively, for FC2. Benaggoune et al. [28] investigated a conditional CNN model and a nonconditional dilated CNN model with attention for multi-step ahead degradation prediction with enhanced lookback and long-

term prognostics, respectively. They reported the median of RMSE and MAPE over 50 trials in the prediction results for 5-, 10-, 15-, and 24-step ahead degradation prediction. The dilated CNN model with attention for 5-step ahead prediction demonstrated the best median RMSE and MAPE of 4.400E-3 and 0.702E-1, respectively, when trained on 591 h of FC1. When trained on 681 h of FC2, the same model with 5-step ahead prediction showed the best RMSE and MAPE of 5.500E-3 and 0.973E-1, respectively. For FC1, the dilated CNN models with attention predicted an RUL of 90 h for an actual RUL of 437 h, so the RUL RE can be computed as 7.940E-1, while for FC2, the models predicted an RUL of 196 h for an actual RUL of 227 h, so the RUL RE can be computed as 1.366E-1. Interestingly, none of these works evaluated the transfer learning performance of their models.

Compared to Refs. [24–26,28], the proposed models exhibit a much smaller RMSE for both FC1 and FC2 and smaller MAPE than in Refs. [24,26,28]. The superior performance of the proposed models is attributed to the intelligent hyperparameter optimization carried out using TPE and hyperband pruning. However, the best RUL prediction RE of the proposed models is at least one order of magnitude smaller than that of the others, except for that in Ref. [26]. Since none of the compared works considered transfer learning, the transfer learning results of the proposed models could not be compared. Hence, the proposed models offer better RUL and long-term degradation prediction performance than the other models.

Conclusion

This work presented a deep learning-based prognostic approach to predict the long-term degradation trend and the remaining useful life (RUL) of five-cell proton exchange membrane fuel cell (PEMFC) stacks under static and dynamic loading. Firstly, the following six deep learning prognostic models were developed using the Keras library in Python: LSTM, 1D–CNN–LSTM, 1D–CNN–Bidirectional-LSTM, GRU, 1D–CNN–GRU, and 1D–CNN–Bidirectional-GRU. The hyperparameters of each model were tuned using the open-source implementations of tree-structured Parzen estimators (TPE) and hyperband pruning in Optuna. The performance of each model was verified on the IEEE PHM 2014 Data Challenge datasets. The prediction results showed that the proposed deep learning architectures could realize more accurate degradation and RUL prediction than most deep learning models reported in the existing literature.

Test results indicated that the best model for FC1 (LSTM) demonstrated the best R^2, RMSE, MAPE, and RE of 98.87E-2, 1.26E-3, 0.285E-1, and 0.0425E-1, respectively. The best model for FC2 (CNN-Bidirectional GRU) showed the best R^2, RMSE, MAPE, and RE of 99.48E-2, 1.77E-3, 0.444E-1, and 0.089E-1, respectively. The best model for FC1 (LSTM) also showed excellent transfer learning results on FC2 with the best R^2, RMSE, MAPE, and RE of 98.91E-2, 0.362E-1, 2.042E-3, and 0.044E-1, respectively. Since the proposed methodology does not assume specific operating modes of the PEMFC, it can be explored for other fuel-cell datasets recorded under various loading conditions. Future studies may also explore ensembles of deep learning models with recurrent and convolutional layers and balance low computation time with high accuracy in RUL and long-term degradation trend prediction. Another research direction could be to extract easily available stack measurements as features for training deep learning models and integrate these models with state estimators for unavailable measurements. Also, the predictive maintenance actions may be classified by autonomous agents based on the online degradation prediction results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviations

CNN	convolutional neural network
GRU	gated recurrent unit
GWO	gray-wolf optimizer
EI	expected improvement
ESN	echo state network
LSTM	long short-term memory
MAE	mean absolute error
MAPE	mean absolute percentage error
PEMFC	proton exchange membrane fuel cell
PHM	prognostics and health management
PSO	particle swarm optimization
RE	relative error
RMSE	root mean square error
RNN	recurrent neural network
RUL	remaining useful life
SGF	Savitzky-Golay filter
TPE	tree-structured Parzen estimators

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